

FIN Local Research — Release 10.44

Checkpoint P471–P473

Exact O167 Primal–Dual Attainment by Polynomial Krawczyk Certification

Krzysztof Żuchowski

Independent Researcher — Fractal Information Theory Project

ORCID: 0009-0002-0909-3613

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Central theorem. A strict rational Krawczyk inclusion proves one exact positive O167 primal–dual contact for the declared reduced three-slot ordered comb. Its global value satisfies

$$0.5233281002710117 \leq D_3^{\text{global}} \leq 0.5233281002710717,$$

an interval of width 6×10^{-14} .

Method. The complete support-ladder condition is reduced symbolically to 13 cubic polynomial residual orbits, enclosed with exact rational arithmetic, and supplemented by exact box-positivity certificates.

Boundary. One exact global optimizer is proved; uniqueness over the complete causal cone remains open. No selector, dimensional source, laboratory record, complete legacy–strict bridge, role transfer, or physical closure is claimed.

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Publication statement

This monograph reports local analytical and computer-assisted Programs P471, P472, and P473. It uses no laboratory record, external audit, internet result, remote computation, or physical validation.

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Chapter 1

Executive summary

This checkpoint completes local Programs P471, P472, and P473. It turns the floating full-KKT evidence of P469 into a finite exact certificate. The proof uses symbolic polynomial reduction, exact rational interval arithmetic, exact trigonometric enclosures, a rational Krawczyk operator, exact Sylvester positivity, and finite-dimensional primal–dual identities. It uses no internet result, external audit, laboratory record, remote computation, or empirical physical evidence.

The principal results are:

1. **P471** — **[Proven]**. The matrix absolute value in the O177 support ladder can be eliminated. For a positive normalizer N and positive dual support X_3 ,

$$X_3 N X_3 = \frac{1}{4} \Delta N \Delta$$

is equivalent to

$$X_3 = \frac{1}{2} N^{-1/2} \left| N^{1/2} \Delta N^{1/2} \right| N^{-1/2}.$$

The exact O167 symmetries and the complete lower-level block-equality KKT conditions reduce X_3 to ten real parameters. Together with the three O167 normalizer parameters (A, B, u) , all 36 upper-triangular Riccati equations collapse symbolically into exactly 13 polynomial residual orbits in 13 unknowns. The maximum polynomial degree is three.

2. **P472** — **[Computer-assisted proof]**. Every one of the 13 polynomials and all 169 Jacobian entries are evaluated with exact rational interval arithmetic. The three trigonometric constants are enclosed by rational intervals at decimal scale 10^{30} . A rational approximate inverse defines the Krawczyk map. Strict inclusions are obtained for radii from 10^{-8} down to 3×10^{-14} . The smaller attempted radius 10^{-14} fails and is not admitted. The selected radius is therefore

$$r = \frac{3}{10^{14}} = 3 \times 10^{-14}.$$

3. **P473** — **[Computer-assisted proof]**. The strict Krawczyk inclusion proves that the exact coefficient instance has one unique root inside the selected box. Separate exact Sylvester calculations prove that the entire box keeps both N and X_3 positive. At this exact root, the Riccati identity proves the two top dual inequalities, the polynomial block pattern makes every lower dual slack zero, and recursive trace identities give equality of the primal and dual objectives. Consequently the O167 normalizer exactly attains the global optimum of the declared reduced three-slot ordered-comb problem.

The exact optimal value is enclosed by

$$0.5233281002710117 \leq D_3^{\text{global}} \leq 0.5233281002710717,$$

an interval of width

$$\frac{3}{5 \times 10^{13}} = 6 \times 10^{-14}.$$

This is an exact attainment theorem, not merely a small primal–dual gap. It proves one exact globally optimal O167 normalizer. It does not yet prove that this optimizer is the only full-cone optimizer outside the certified local root box.

The new objects are O179 Polynomialized Interior KKT Core, O180 Rational Polynomial Krawczyk Provider, and O181 Exact O167 Primal–Dual Contact Certificate.

No FIN selector, QW-2191 discharge, dimensional source, physical clock, preparation, apparatus, independent experimental record, complete legacy-to-strict kernel bridge, legacy role transfer, L_{total} , Standard Model or gravity closure, or Theory of Everything is claimed.

Chapter 2

Confidence convention

Label	Meaning
[Proven]	Exact analytic or finite symbolic consequence of the declared premises
[Computer-assisted proof]	Finite rational or outward-interval certificate with auditable artifacts
[Strong evidence]	Reproducible numerical evidence lacking a complete exact certificate
[Conditional]	Conclusion dependent on an explicitly supplied mathematical interface
[Open]	Current local artifacts do not decide the proposition
[Refuted]	A stated candidate or attempted certificate fails a necessary test
[Blocked by external evidence]	Laboratory, calibration, custody, or an external record is indispensable

Chapter 3

State transition and guardrails

Lane	Release 10.43 state	Release 10.44 state
O167 stationarity	Full floating KKT residual 1.33×10^{-15}	Exact polynomial root exists in a certified box
O167 global attainment	Open	Proved for the declared reduced three-slot problem
Global value	Rational bracket $4.1055151 \times 10^{-10}$	width Exact-attainment value interval width 6×10^{-14}
Root uniqueness	Numerical locator	Unique exact polynomial root in the selected local box
Full-cone optimizer uniqueness	Open	Still open outside the local root box
Selector / QW-2191	Open	Unchanged
Dimensional source	Missing	Unchanged
Laboratory evidence	None	None
Legacy-to-strict bridge	Incomplete	Unchanged

The intermediate and strict kernels remain separate:

$$K_{\text{legacy,ont}}(d) = \frac{\alpha_{\text{geo}} \cos(\omega d + \phi)}{1 + \beta_{\text{tors}} d}, \quad K_{\text{strict,gate}}(d) = \frac{\cos(\omega d + \phi)}{1 + \beta d^n}.$$

The present theorem is downstream of supplied finite channel operators. It derives neither kernel, completes no bridge, and transfers no legacy physical role. The ontology remains

$$\text{nadsoliton} \longrightarrow \text{light} \longrightarrow \text{matter} \longrightarrow \text{emergent observer},$$

with no informational substrate below the nadsoliton.

Chapter 4

Inherited problem

4.1 Three-slot primal

For the supplied compressed Hermitian process difference Δ , the ordered three-slot primal SDP is

$$\begin{aligned} & \text{maximize} && \frac{1}{2} \text{Tr}[\Delta(T_+ - T_-)] \\ & \text{subject to} && T_+ + T_- = B_0 \oplus B_1, \\ & && B_0 + B_1 = C_0 \oplus C_1, \\ & && C_0 + C_1 = \text{diag}(d_0, d_1), \\ & && d_0 + d_1 = 1, \end{aligned}$$

with all primal variables positive semidefinite.

For a fixed positive normalizer $N = T_+ + T_-$,

$$\max_{T_{\pm}} \frac{1}{2} \text{Tr}[\Delta(T_+ - T_-)] = \frac{1}{2} \left\| N^{1/2} \Delta N^{1/2} \right\|_1.$$

4.2 Nested dual

The dual minimizes λ subject to

$$X_3 \succeq \pm \frac{\Delta}{2},$$

$$X_2 \succeq (X_3)_{00}, \quad X_2 \succeq (X_3)_{11},$$

$$X_1 \succeq (X_2)_{00}, \quad X_1 \succeq (X_2)_{11},$$

$$\lambda \geq (X_1)_{00}, \quad \lambda \geq (X_1)_{11}.$$

Slater points exist on both sides. Therefore finite-dimensional strong duality applies.

4.3 O167 normalizer

The O167 normalizer is parameterized by

$$N(A, B, u) = B_0 \oplus B_1, \quad C = \frac{1}{2} - A - 2B,$$

$$B_0 = \begin{pmatrix} A & u & u & 0 \\ u & B & 0 & -u \\ u & 0 & B & -u \\ 0 & -u & -u & C \end{pmatrix},$$

$$B_1 = \begin{pmatrix} C & -u & -u & 0 \\ -u & B & 0 & u \\ -u & 0 & B & u \\ 0 & u & u & A \end{pmatrix}.$$

This form satisfies the causal trace recursion identically whenever the blocks are positive.

Chapter 5

P471 — polynomialization of the support ladder

5.1 Removing the matrix absolute value

Set

$$H = N^{1/2}\Delta N^{1/2}.$$

The P469 support formula is

$$X_3 = \frac{1}{2}N^{-1/2}|H|N^{-1/2}.$$

Multiplying through gives

$$X_3NX_3 = \frac{1}{4}N^{-1/2}|H|^2N^{-1/2} = \frac{1}{4}\Delta N\Delta.$$

The reverse implication is equally important. Suppose $N \succ 0$, $X_3 \succ 0$, and

$$X_3NX_3 = \frac{1}{4}\Delta N\Delta.$$

Then

$$Y = N^{1/2}X_3N^{1/2} \succ 0$$

obeys

$$Y^2 = \frac{1}{4}H^2.$$

The positive square root of $H^2/4$ is unique. Hence

$$Y = \frac{1}{2}|H|$$

and the support formula follows. Thus the nonsmooth-looking matrix absolute value is exactly equivalent, inside the positive cone, to a finite cubic polynomial system.

5.2 Exact lower-ladder pattern

At an interior optimum, positive $B_0, B_1, C_0, C_1, d_0, d_1$ force both lower dual slacks at every level to vanish. Therefore the two diagonal blocks of X_3 equal one common X_2 , the two diagonal blocks of X_2 equal one common X_1 , and the two diagonal elements of X_1 equal λ .

Under the exact O167 permutation and complement pattern,

$$X_2 = \begin{pmatrix} L & a & a & c \\ a & L & b & a \\ a & b & L & a \\ c & a & a & L \end{pmatrix},$$

and

$$X_3 = \begin{pmatrix} X_2 & Y \\ Y^T & X_2 \end{pmatrix},$$

where

$$Y = \begin{pmatrix} d & e & e & f \\ g & h & h & e \\ g & h & h & e \\ i & g & g & d \end{pmatrix}.$$

The unknown vector is

$$z = (A, B, u, L, a, b, c, d, e, f, g, h, i) \in \mathbb{R}^{13}.$$

5.3 Exact residual orbits

Expanding

$$R(z) = X_3(z)N(z)X_3(z) - \frac{1}{4}\Delta N(z)\Delta$$

gives 36 upper-triangular entries. Exact symbolic comparison over $\mathbb{Q}[s_1, s_2, s_3]$, where

$$s_k = \sin \frac{k\pi}{8},$$

partitions those entries into exactly 13 identical polynomial orbits:

Orbit	Upper-triangular positions
1	(0,0), (7,7)
2	(0,1), (0,2), (5,7), (6,7)
3	(0,3), (4,7)
4	(0,4), (3,7)
5	(0,5), (0,6), (1,7), (2,7)
6	(0,7)

Orbit	Upper-triangular positions
7	(1,1), (2,2), (5,5), (6,6)
8	(1,2), (5,6)
9	(1,3), (2,3), (4,5), (4,6)
10	(1,4), (2,4), (3,5), (3,6)
11	(1,5), (1,6), (2,5), (2,6)
12	(3,3), (4,4)
13	(3,4)

Selecting one representative from each orbit yields a square 13-variable system $F(z) = 0$. Its maximum degree is three. No equation is discarded: the orbit identities prove that the 13 representatives vanish if and only if the complete symmetric matrix residual vanishes.

5.4 Floating locator and necessity boundary

The local numerical root is

$$(A, B, u) \approx (0.28716641537176957, \\ 0.06877721090495262, \\ 0.009493709596686299),$$

with

$$L \approx 0.5233281002710417.$$

The full 13-vector residual infinity norm is

$$6.158268339717665 \times 10^{-17}.$$

The Jacobian singular values range from

$$3.833325700590803$$

to

$$0.003267495644871251,$$

giving condition number approximately 1173.17. This is sufficient for a well-scaled local interval attempt, but P471 alone does not convert a floating locator into an exact root.

Chapter 6

P472 — exact rational interval provider

6.1 Arithmetic layer

Each polynomial is stored as a finite list of rational coefficients and monomials. Every primitive operation uses an exact interval

$$[x_-, x_+], \quad x_{\pm} \in \mathbb{Q}.$$

Addition and multiplication are performed by rational endpoint arithmetic. No binary floating number appears in the proof evaluator.

The exact Taylor/Machin trigonometric providers are coarsened outward to denominator 10^{30} . Their original widths are much smaller; for example the inherited enclosure of $\sin(\pi/8)$ has width below 7×10^{-67} . Coarsening reduces integer growth while preserving the true constants.

6.2 Krawczyk operator

Let $z_0 \in \mathbb{Q}^{13}$ be the rationalized locator, Z a rational box centered at z_0 , and $C \in \mathbb{Q}^{13 \times 13}$ a rationalized inverse of the midpoint Jacobian. The exact interval Krawczyk map is

$$K(z_0, Z) = z_0 - CF(z_0) + [I - CF'(Z)](Z - z_0).$$

If

$$K(z_0, Z) \subset \text{int } Z,$$

then the exact coefficient instance has one root in Z , and that root is unique within Z .

6.3 Attempt ladder

The following radii were tested with the same exact evaluator:

Radius	Result
10^{-8}	strict inclusion
10^{-9}	strict inclusion
10^{-10}	strict inclusion
10^{-11}	strict inclusion
10^{-12}	strict inclusion
10^{-13}	strict inclusion
3×10^{-14}	strict inclusion
10^{-14}	rejected

The selected radius is the smallest admitted attempted radius:

$$r = 3 \times 10^{-14}.$$

The strict inward margin is positive. The rejected 10^{-14} attempt is retained in the CSV artifact and is not silently rounded into success.

6.4 What the provider proves

P472 encloses the complete polynomial residual and every Jacobian entry on the entire box. It does not sample corners and does not assume that a matrix eigendecomposition is interval stable. The proof layer therefore removes the principal methodological weakness identified in P469.

Chapter 7

P473 — exact O167 primal–dual attainment

7.1 Exact root

The strict P472 inclusion proves the existence of an exact root

$$z_* \in z_0 + [-r, r]^{13}, \quad r = 3 \times 10^{-14}.$$

The root is unique within this box. The global-value coordinate L_* obeys

$$\frac{5233281002710117}{10^{16}} \leq L_* \leq \frac{5233281002710717}{10^{16}}.$$

7.2 Positivity on the entire box

Krawczyk inclusion alone does not establish that the root lies in the positive primal–dual cone. P473 separately rationalizes the center matrices and applies exact Sylvester tests.

For the normalizer center,

$$N(z_0) \succeq \frac{1}{100}I.$$

Every normalizer entry moves by at most $5r$, and therefore

$$\|\delta N\|_2 \leq 8(5r) = 40r.$$

The certified box lower bound is

$$\lambda_{\min}(N(z)) \geq \frac{24999999997}{2500000000000} > 0.$$

For the dual support center,

$$X_3(z_0) \succeq \frac{1}{100}I,$$

and

$$\|\delta X_3\|_2 \leq 8r.$$

Thus

$$\lambda_{\min}(X_3(z)) \geq \frac{124999999997}{12500000000000} > 0.$$

Both matrices are positive throughout the complete certified box, hence also at the exact root.

7.3 Exact top dual feasibility

At the root,

$$X_3 N X_3 = \frac{1}{4} \Delta N \Delta.$$

Set

$$H = N^{1/2} \Delta N^{1/2}, \quad Y = N^{1/2} X_3 N^{1/2}.$$

Since $Y \succ 0$ and $Y^2 = H^2/4$,

$$Y = \frac{1}{2}|H|.$$

For every Hermitian H ,

$$|H| \succeq \pm H.$$

Congruence by $N^{-1/2}$ gives

$$\boxed{X_3 \succeq \pm \frac{\Delta}{2}.}$$

The two top dual inequalities are therefore exact consequences of the polynomial root and positivity; they are not numerical eigenvalue tests.

7.4 Exact lower dual feasibility

The ten-parameter pattern makes the two diagonal blocks of X_3 equal to the same X_2 . The two diagonal blocks of X_2 equal the same

$$X_1 = \begin{pmatrix} L & a \\ a & L \end{pmatrix},$$

and both diagonal entries of X_1 equal L . Taking

$$\lambda = L$$

makes all six lower dual slacks exactly zero. Zero is positive semidefinite, so the complete dual ladder is feasible.

7.5 Exact equality of primal and dual values

The trace-norm support identity gives the fixed-normalizer primal value

$$D(N) = \text{Tr}(NX_3).$$

The block identities telescope:

$$\begin{aligned} \text{Tr}(NX_3) &= \text{Tr}[(B_0 + B_1)X_2] \\ &= \text{Tr}[(C_0 + C_1)X_1] \\ &= \text{Tr}[\text{diag}(d_0, d_1)X_1] \\ &= L(d_0 + d_1) \\ &= L = \lambda. \end{aligned}$$

Thus a feasible primal construction and a feasible dual construction have the same objective value. Weak duality alone proves that both are globally optimal.

7.6 Theorem

Exact O167 attainment theorem. [Computer-assisted proof] For the exact declared reduced three-slot phase/dephasing channel at $q = 4/5$ and $\theta = \pi/8$, there exists an exact positive O167 normalizer whose binary causal discrimination value equals the global full-cone optimum. The optimal value lies in

$$[0.5233281002710117, 0.5233281002710717].$$

The exact root is unique in the certified 13-dimensional box of radius 3×10^{-14} .

The theorem proves existence and exact global attainment of one O167 optimizer. It does not prove that no disconnected or distant globally optimal normalizer exists elsewhere in the 21-dimensional causal cone.

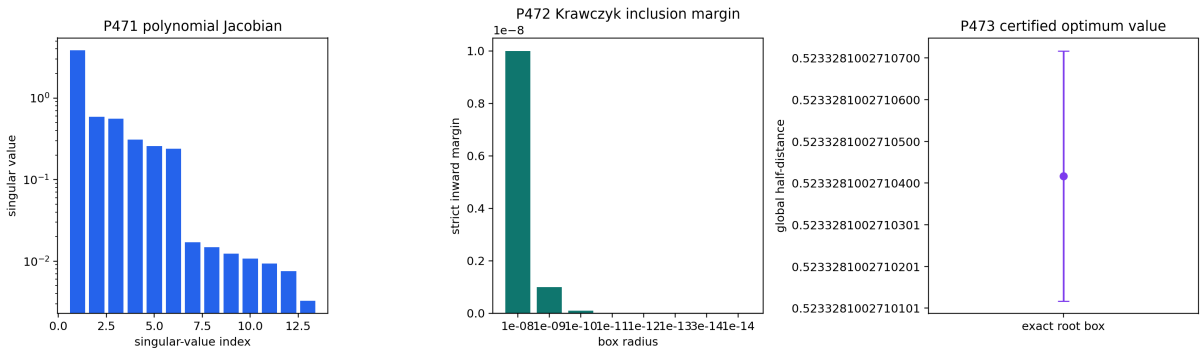


Figure 7.1: Polynomial Jacobian spectrum, exact Krawczyk inclusion margins, and the certified global optimum interval.

Chapter 8

New objects

8.1 O179 — Polynomialized Interior KKT Core

Field	Definition
Input	O167 normalizer, structured nested dual, supplied Δ
Equation	$X_3 N X_3 = \Delta N \Delta / 4$
Unknown count	13
Complete residual count	36 upper-triangular entries
Exact orbit count	13
Maximum degree	3
Generated theorem	Positive roots are exactly structured support-ladder contacts
Confidence	[Proven]

8.2 O180 — Rational Polynomial Krawczyk Provider

Field	Definition
Coefficients	Exact rational polynomial terms plus outward rational sine boxes
Jacobian	All 169 entries enclosed exactly
Preconditioner	Rationalized inverse of the midpoint Jacobian
Admission	Strict coordinatewise inclusion $K(Z) \subset \text{int } Z$
Selected box radius	3×10^{-14}
Failure retained	10^{-14} box rejected
Confidence	[Computer-assisted proof]

8.3 O181 — Exact O167 Primal–Dual Contact Certificate

Field	Definition
Root	Unique exact O179 root in the O180 box
Positivity	Exact Sylvester margins for N and X_3 , perturbation paid
Top dual	Riccati identity plus positive-square-root uniqueness
Lower dual	Exact zero slacks from nested block equality
Equality	Recursive trace telescope gives primal = λ
Generated theorem	Exact global O167 attainment
Remaining open	Full-cone optimizer uniqueness outside the root box
Confidence	[Computer-assisted proof]

Chapter 9

Falsification and independent checks

9.1 Algebraic checks

- All 36 upper-triangular matrix equations are retained through exact orbit identities.
- Sine constants remain symbolic during orbit classification.
- The Riccati implication is used only after positivity is independently certified.
- The lower dual pattern is written explicitly; no floating near-equality is promoted.

9.2 Interval checks

- Polynomial evaluation and Jacobian evaluation use exact fractions.
- Trigonometric enclosures are outward, not nearest rounded.
- The preconditioner is rationalized.
- Inclusion is checked coordinate by coordinate with a strictly positive margin.
- A smaller failed box is recorded as failed.
- The entire root box, not only its center, is proved positive.

9.3 Duality checks

- Top feasibility is derived from $|H| \succeq \pm H$.
- Lower feasibility uses exact zero slacks.
- The primal value is linked to the dual scalar by an explicit trace telescope.
- Strong duality is not needed for the final equality; feasible primal and dual points with the same value plus weak duality suffice.

9.4 Claims rejected

The following are not consequences of P473:

- the full ordered-comb optimizer is globally unique;
- O167 is universally optimal for other channels or slot counts;
- a causal tester is a FIN strict selector;
- L_* is a dimensional action, time, length, mass, or energy;
- a finite operator theorem supplies a physical preparation or detector;
- the legacy kernel inherits strict-kernel physical roles;
- a local mathematical optimum is empirical confirmation.

Chapter 10

Significance for FIN

Release 10.44 closes a well-defined mathematical problem that had remained open since P451. The progression is now:

coherent primal witness $\longrightarrow$ nested rational dual bounds $\longrightarrow$ full floating KKT locator
 $\longrightarrow$ polynomial exact root $\longrightarrow$ exact primal–dual contact.

This is a genuine upgrade from numerical plausibility to a finite theorem. It demonstrates that the finite operator layer can support proof-grade causal optimization, including coherence and memory, without relying on an external SDP solver.

The result remains a theorem about a supplied dimensionless channel. It does not bridge information to SI units or laboratory observables. It does not select a physical branch or origin. It does not complete the legacy-to-strict kernel map. The missing physics objects identified by earlier audits remain missing.

Chapter 11

Ranked next research programs

Rank	Program	Question	Method and success criterion
1	P474	Is the P473 globally optimal normalizer unique on the complete causal cone?	Exact tangent-cone nondegeneracy, top-slack kernel analysis, and strict-complementarity uniqueness theorem or counterexample
2	P483	Does the exact root persist under a certified (q, θ) parameter neighborhood?	Parametric interval implicit-function/Krawczyk tube
3	P475	Can the optimal value or root receive an algebraic minimal polynomial?	Elimination over the algebraic sine field; stop if degree/height is computationally prohibitive
4	P479	Formalize the Riccati-to-support and trace-telescope theorem	Lean finite-matrix theorem with positivity premises separated
5	P477	Does O179 generalize to arbitrary slot number?	Inductive nested support Riccati ladder and symmetry-orbit count
6	P478	Does an exact coherent optimum exist for four slots?	Reduced four-slot primal, structured dual, rational witness before any global claim
7	P476	Extend P465 uniqueness over a (q, θ) region	Exact condition $4q^2 \cos^2 \theta > 1$ plus positive sector-weight boundary audit
8	P480	Build a standalone minimal checker for O180/O181	Serialize rational monomials, boxes, and Krawczyk matrix; independent replay without SciPy
9	P481	Classify all roots of the 13-polynomial system in the positive feasible parameter domain	Interval branch-and-bound or certified exclusion boxes
10	P460	Test O174 allocation stability under alternative admissible loss geometries	Quotient sensitivity and ranking-margin theorem
11	P470	Audit local operational conclusions under strict-kernel parameter boxes	Interval perturbation with no legacy role transfer
12	P463	Specify the minimal preparation/instrument contract	Typed conditional interface, explicitly non-empirical

The next bounded batch is

$$P474 \longrightarrow P483 \longrightarrow P479.$$

P474 has highest priority because P473 proves one global optimizer but does not yet classify the complete optimal face. P483 tests robustness rather than another isolated parameter point. P479 isolates the reusable exact theorem from its numerical certificate.

Chapter 12

Artifacts and restart

- FIN_Programs_471_472_473_Results.json
- FIN_Programs_471_472_473_Summary.csv
- FIN_Programs_471_472_473_P471_Polynomial_Core.csv
- FIN_Programs_471_472_473_P472_Interval_Provider.csv
- FIN_Programs_471_472_473_P473_Krawczyk.csv
- FIN_Programs_471_472_473_P473_Root_Box.npz
- FIN_Programs_471_472_473_Figures/p471_p473_polynomial_krawczyk.png
- fin_programs_471_472_473.py
- test_fin_programs_471_472_473.py

Recompute with:

```
MPLCONFIGDIR=/tmp/fin-mpl-1044 python3 fin_programs_471_472_473.py
python3 -m unittest -v test_fin_programs_471_472_473.py
python3 fin_programs_471_472_473_to_latex.py
lualatex -interaction=nonstopmode -halt-on-error \
    FIN_Local_Research_Checkpoint_P471_P473_EN.tex
sha256sum -c FIN_PROGRAMS_471_472_473_RELEASE_MANIFEST.sha256
```

Chapter 13

Conclusion

P471–P473 close the exact-attainment question for the declared reduced three-slot ordered-comb optimization. The key move is conceptual as well as computational: the spectral support map is replaced by a positive Riccati equation, exposing a 13-variable cubic core. Exact rational Krawczyk inclusion then proves a true root, and positivity plus trace recursion turns that root into matching primal and dual feasible points.

The resulting theorem is strong but narrow. One exact O167 optimizer exists and globally attains a value in an interval only 6×10^{-14} wide. Whether it is the only optimizer in the entire cone remains open. The result advances FIN’s finite operator and information-processing mathematics; it does not manufacture the selector, dimensions, preparation, apparatus, or experimental record required for physics.

Reproduction certificate

```
MPLCONFIGDIR=/tmp/fin-mpl-1044 \  
python3 fin_programs_471_472_473.py  
python3 -m unittest -v \  
    test_fin_programs_471_472_473.py \  
    test_fin_checkpoint_p471_p473.py  
python3 fin_programs_471_472_473_to_latex.py  
lualatex -interaction=nonstopmode -halt-on-error \  
    FIN_Local_Research_Checkpoint_P471_P473_EN.tex  
sha256sum -c FIN_PROGRAMS_471_472_473_RELEASE_MANIFEST.sha256
```

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